SAP CAPSTONE PROJECT

Procure-to-Pay (P2P) — Full Purchasing Cycle using SAP MM

Name:	Pravin Kumar
Roll No:	23051688
Branch:	CSE (B.Tech)
Submission Date:	21 April 2026

Title

Procure-to-Pay (P2P): A Comprehensive Study of the End-to-End Procurement Workflow Implemented through SAP Materials Management (MM)

Problem Statement

Across a wide range of enterprises, the procurement function is either carried out through entirely manual workflows or through isolated, non-integrated software platforms. Such an approach gives rise to a series of operational challenges: purchase approvals are delayed, coordination with suppliers becomes inconsistent, real-time stock visibility is absent, and vendor invoices are frequently processed with errors. Taken together, these shortcomings drive up costs and diminish overall business efficiency.

To address these issues, organisations require a unified, process-driven platform capable of governing the entire procurement cycle — from initial requisition through to final payment — within a single, automated environment.

Procure-to-Pay (P2P) Process Overview

Explanation of Flow

The P2P workflow follows a well-defined sequence of procurement activities, each building upon the previous stage:

- **1. Purchase Requisition (PR)** — An internal department raises a formal request specifying the required goods or services.
- **2. Purchase Order (PO)** — The requisition is converted into a binding purchase order and dispatched to the selected supplier.
- **3. Goods Receipt (GR)** — Upon delivery, materials are physically inspected and formally acknowledged in the system.
- **4. Invoice Verification (IV)** — The supplier's invoice undergoes a three-way check against the PO and GR to confirm accuracy.
- **5. Payment** — Once verified, the financial module releases the payment to the vendor as per agreed terms.

SAP T-Codes Table

Step	Activity	SAP T-Code
1	Create Purchase Requisition	ME51N
2	Create Purchase Order	ME21N
3	Goods Receipt	MIGO
4	Invoice Verification	MIRO
5	Payment	F110

Process Flow Diagrams

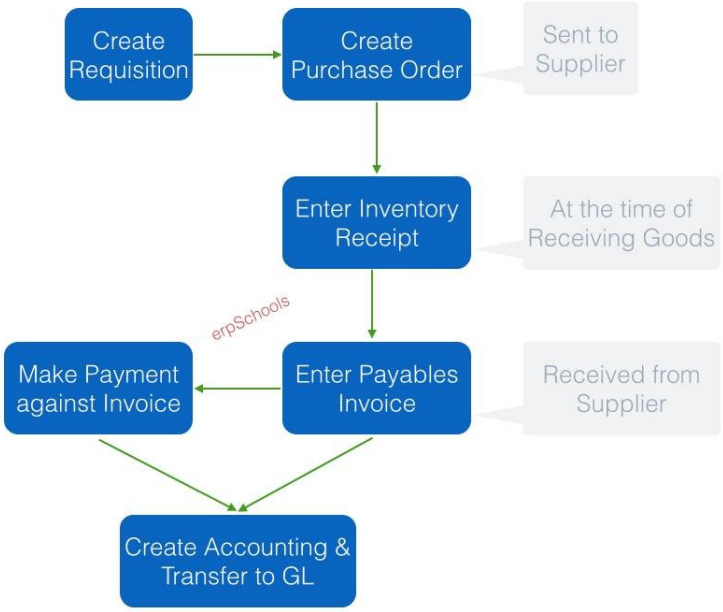


Fig. 1: Procure-to-Pay Basic Flow – Requisition to GL Transfer

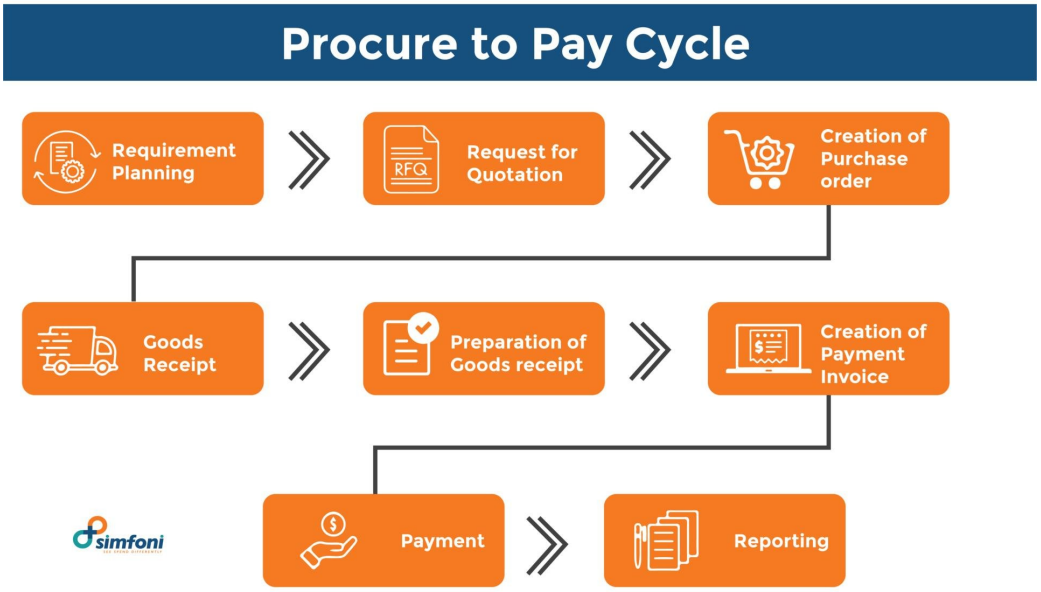


Fig. 2: Procure-to-Pay Cycle – End-to-End Overview

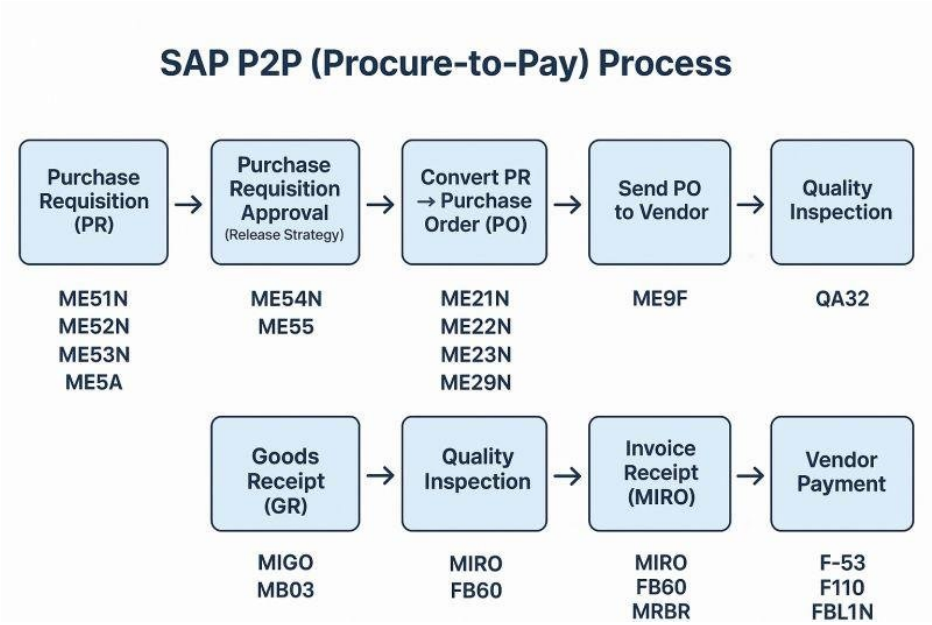


Fig. 3: SAP P2P Process with T-Codes

Fig. 4: Procure-to-Pay Process – Six Key Stages

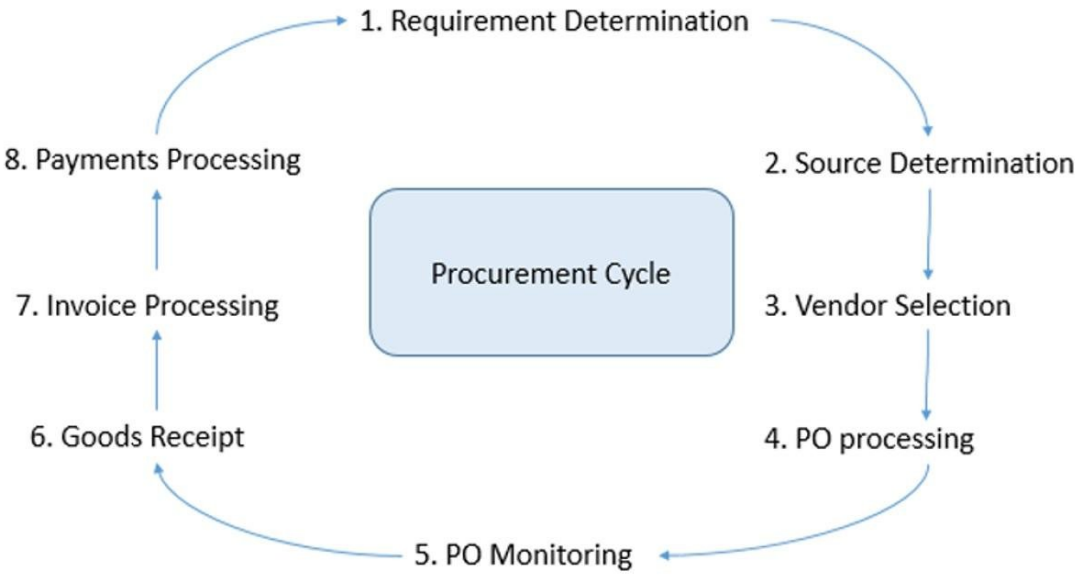


Fig. 5: Procurement Cycle – Eight-Step Circular Flow

Detailed Process Explanation

1. Purchase Requisition

A department user submits a requisition within the SAP system, specifying the item description, quantity needed, and the required delivery date. This triggers the formal procurement workflow.

2. Purchase Order

The approved requisition is transformed into a purchase order, which captures agreed unit prices, payment conditions, and expected delivery timelines before being transmitted electronically to the vendor.

3. Goods Receipt

When the vendor delivers the goods, the warehouse team records the receipt in SAP using T-code MIGO. Stock levels are updated instantly, and a material document is generated to maintain a full audit trail.

4. Invoice Verification

The incoming supplier invoice is entered via MIRO and validated through a three-way matching process — cross-referencing the purchase order, the goods receipt document, and the invoice — to detect and prevent billing discrepancies.

5. Payment Processing

Following successful invoice verification, the SAP FI module automatically schedules and executes the vendor payment run, ensuring timely settlement and accurate ledger reconciliation.

System Architecture / Integration Diagrams

The following diagrams present the technical integration between SAP MM and SAP FI, demonstrating how procurement events in the MM module automatically trigger corresponding financial postings in real time:

- SAP MM governs all purchasing operations — covering requisitions, purchase orders, goods receipts, and invoice capture
- SAP FI manages the downstream financial impact — including GL postings, accounts payable entries, payment runs, and reconciliation
- Seamless integration between the two modules guarantees that accounting records remain current at every transaction milestone

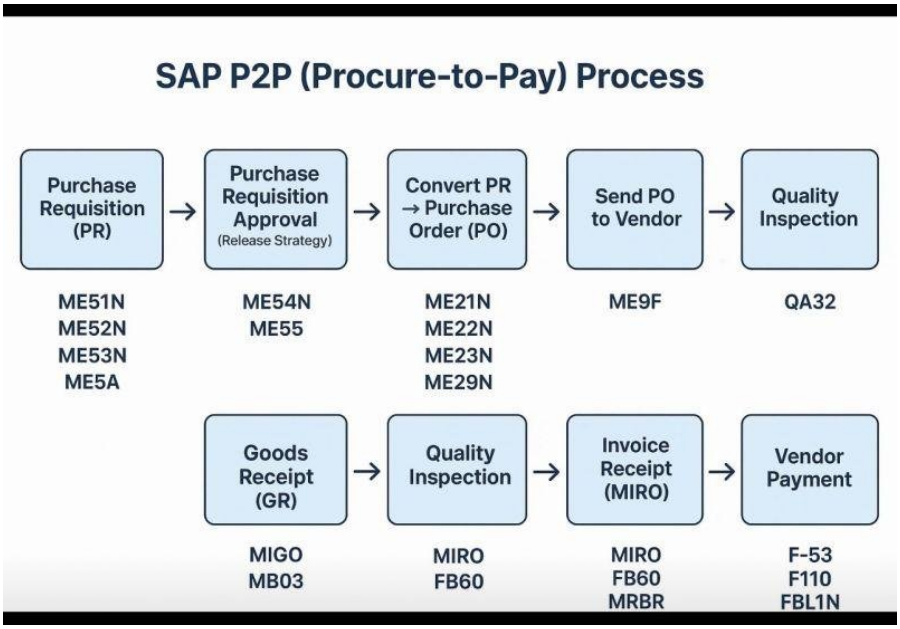


Fig. 6: SAP P2P Detailed Process Flow

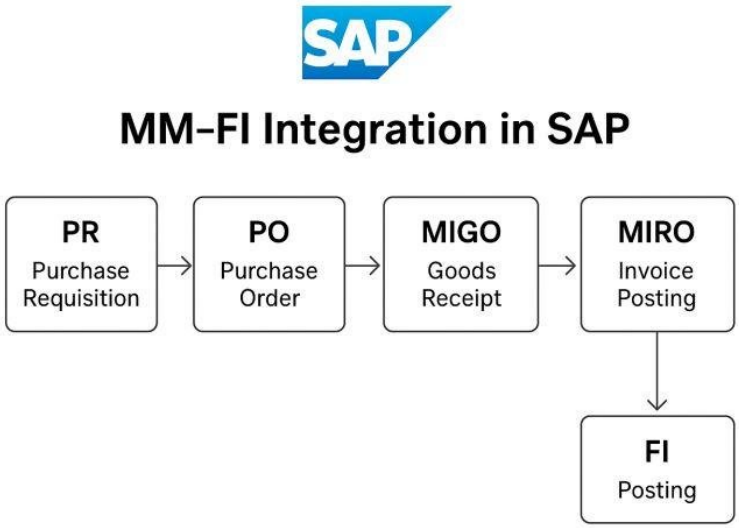


Fig. 7: MM-FI Integration in SAP (PR → PO → MIGO → MIRO → FI)

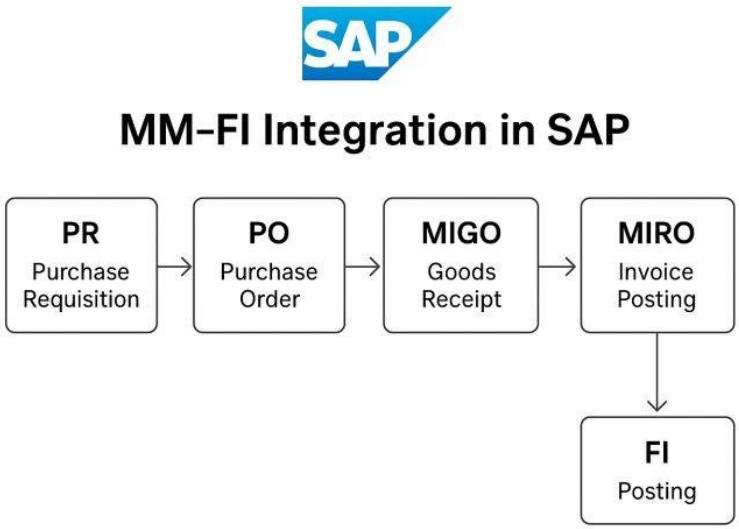


Fig. 8: MM-FI Integration Flow Diagram

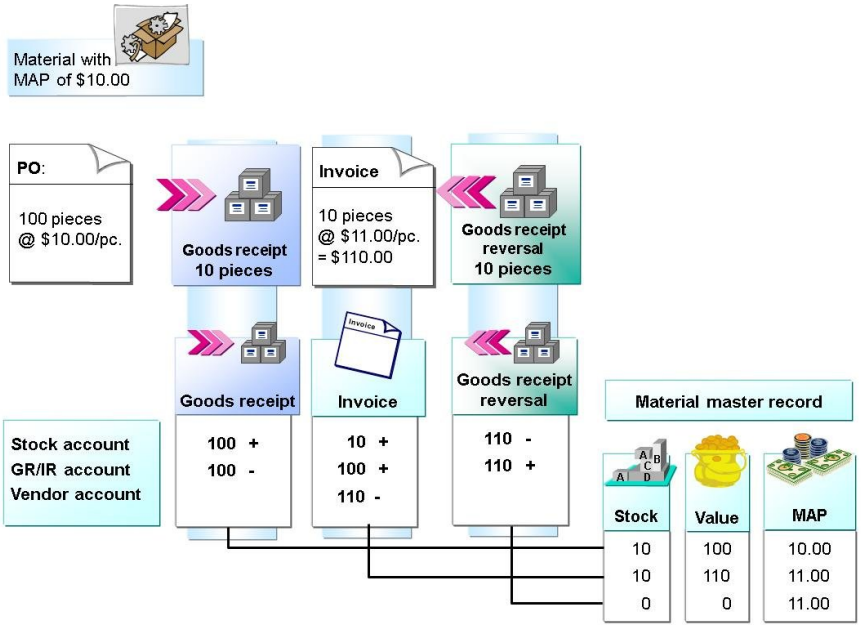


Fig. 9: Moving Average Price (MAP) and GR/IR Accounting

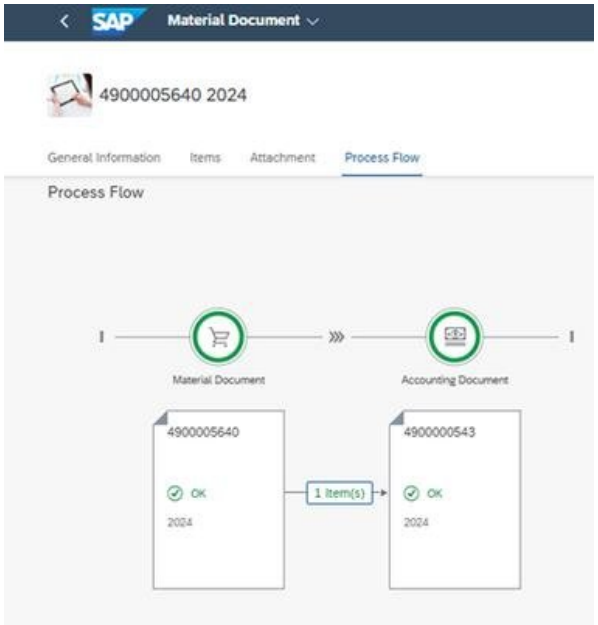


Fig. 10: Material Document and Accounting Document Process Flow

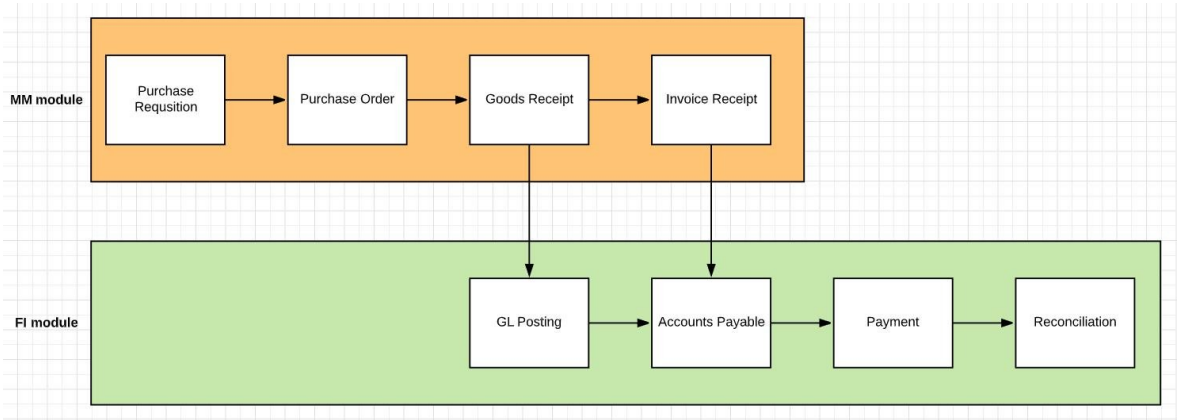
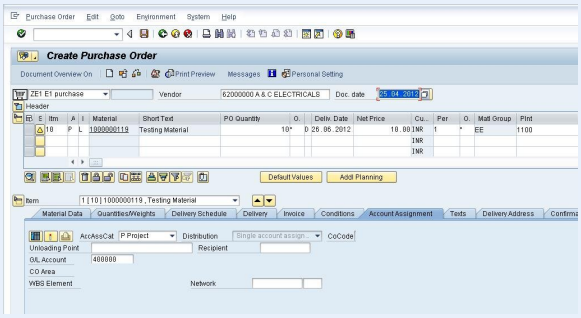


Fig. 11: SAP MM Module to FI Module Integration Architecture

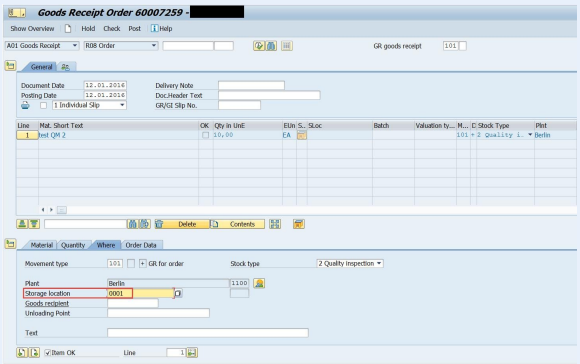
Screenshots

The screens captured below represent the live SAP GUI environments navigated during the execution of the Procure-to-Pay project:

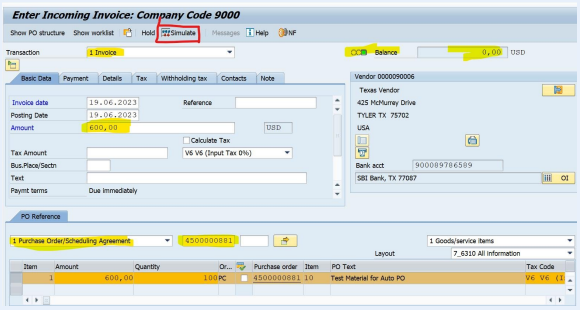
Purchase Order Screen (ME21N)



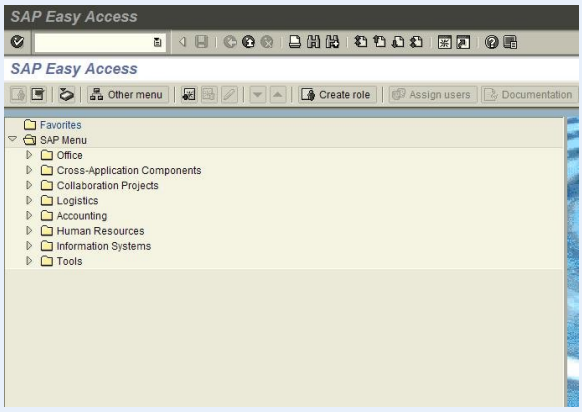
Goods Receipt Screen (MIGO)



Invoice Verification Screen (MIRO)



SAP Dashboard / SAP Easy Access



Tech Stack

Technology	Description
SAP S/4HANA / ECC	ERP Platform
SAP MM	Procurement Module
SAP FI	Financial Module
SAP GUI	User Interface
SAP Database	Data Storage

Unique Points

- Covers the full spectrum of the procurement lifecycle from requisition to payment
- Inventory records are refreshed instantaneously upon each goods receipt posting
- Tight coupling between SAP MM and SAP FI eliminates the need for manual journal entries
- Automated three-way matching significantly lowers the risk of invoice fraud and human error
- Provides end-to-end visibility across all procurement transactions for better decision-making

Future Improvements

- Deployment of machine-learning models to assist in intelligent vendor evaluation and selection
- Full automation of purchase order creation using demand-forecasting algorithms
- Development of a mobile-first interface enabling managers to approve requisitions on the go
- Integration of a real-time analytics dashboard to track spend patterns and procurement KPIs
- Direct API-level connectivity with external supplier portals and e-procurement marketplaces

Conclusion

Implementing the Procure-to-Pay cycle within SAP MM delivers a disciplined and technology-driven framework for managing all purchasing activities. The seamless alignment of procurement, inventory, and financial workflows within a single integrated platform ensures that transactions are accurate, auditable, and completed without unnecessary delay.

By replacing fragmented manual processes with a coherent automated system, organisations can significantly cut operational overhead, reduce error rates, and strengthen inter-departmental collaboration — positioning SAP P2P as a critical enabler of modern, scalable enterprise operations.

Declaration

I hereby affirm that the content presented in this project report is entirely my own academic work. No portion of this submission has been reproduced or adapted from the work of peers, seniors, or any online repository.

Name: Pravin Kumar | **Roll No:** 23051688 | **Branch:** B.Tech CSE | **Date:** 21 April 2026